ModelIt!

Student Activity Pack

Why Do Animals Have Superpowers?

The Science of Body Structures and Survival

Grade 4 | NGSS 4-LS1-1 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What animal do you think has the coolest 'superpower'? What does it do?

2 Why do you think polar bears have thick fur but lizards do not?

3 Draw an animal and label the body parts that help it survive.

4 What do you think would happen if you put a fish on dry land?

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Body Structure Type | Environment Danger Level | Survival Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Perfect Match

Thick fur in arctic cold

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Mismatch

Thick fur in desert heat

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: High Danger

Maximum danger, comparing structures

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Food Availability — How much food is available in the environment — animals need structures to find and eat the right food

Predator Speed — How fast the predators in the environment can move — prey need escape structures to survive

Temperature Extreme — How hot or cold the environment gets — animals need temperature-regulation structures to survive

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

How do animals survive in dangerous environments using just the body parts they were born with? Use evidence from your simulation to support your answers.

1 Why would thick fur be a superpower in the Arctic but a problem in the desert?

2 Can you think of a human-made invention that copies an animal's survival structure?

3 What body structures do YOU have that help you survive?

4 Why do animals in the same environment often have similar body features?

5 How did your model help you understand why animals look the way they do?

8 STEM Challenge: Design a Survival Animal

ModelIt!

YOUR ENGINEERING MISSION

Design an imaginary animal with the perfect body structures to survive in a specific extreme environment using evidence from your model.

THE SCENARIO

A nature documentary team is filming in an extreme environment. They have hired your team to predict what kind of animal could survive there based on the dangers present. Design an animal with body structures that perfectly match the environment's challenges.

Engineering Design Thinking

1. WHAT EXTREME ENVIRONMENT WILL YOUR ANIMAL LIVE IN (DESERT, DEEP OCEAN, ARCTIC, JUNGLE CANOPY)?

2. WHAT EXTERNAL STRUCTURES WILL IT HAVE (FUR, CLAWS, WINGS, CAMOUFLAGE, ARMOR)?

3. WHAT INTERNAL STRUCTURES WILL HELP IT SURVIVE (STRONG BONES, LARGE LUNGS, FAST MUSCLES)?

4. HOW WILL YOU SHOW THAT EACH STRUCTURE MATCHES A SPECIFIC DANGER IN THE ENVIRONMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Wildlife Biologists and Zoologists study how animals survive in the wild and work to protect endangered species and their habitats. They earn $50,000–$85,000/year.